

NITHWIN V M

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SUMMARY

Computer Science and Engineering graduate with hands-on experience in Generative AI, computer vision, and full-stack development. Built LegalPilot AI, a RAG-based legal document assistant, and ShadowCoders, an examination platform serving 400+ students. Solved 750+ LeetCode problems and won two 1st Place university innovation awards for AI-driven solutions.

EDUCATION

Nandha Engineering College

Bachelor of Engineering in Computer Science and Engineering
CGPA: 8.2/10

Erode, TN

2022 – 2026

TECHNICAL SKILLS

- **Languages:** C++, Java, Python, TypeScript, JavaScript, SQL
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming
- **Backend:** Java Spring Boot, Node.js, Express.js, RESTful APIs, WebSockets, Hibernate/JPA
- **Databases:** PostgreSQL, SQLite, Prisma ORM
- **AI & Generative AI:** LangChain, RAG, Gemini API, Groq API, FAISS, Custom YOLOv8, InsightFace
- **Frontend:** Next.js, React.js, Tailwind CSS
- **Developer Tools:** Linux, Git/GitHub, Postman, Vercel

PROJECTS

LegalPilot AI - Legal Document Intelligence Platform | github.com/Nithwin/LegalPilotAI

- Built a RAG-based legal assistant enabling users to query legal documents with grounded responses and page-level citations using Gemini and Groq APIs.
- Developed document processing pipelines using LangChain and FAISS for semantic retrieval, clause extraction, and contract summarization.
- Designed a full-stack application with FastAPI and Next.js supporting legal Q&A, risk analysis, and multi-provider LLM integration.
- **Tech Stack:** FastAPI, Next.js, LangChain, Gemini API, Groq API, FAISS, PyMuPDF, TypeScript

ShadowCoders - Scalable Examination Platform | github.com/Nithwin/ShadowCoders

- Deployed a full-stack exam platform serving 400+ students, maintaining 100% uptime during peak loads of 80+ concurrent users.
- Reduced backend load by 90% via client-side code execution and engineered 11 custom browser-based anti-cheating mechanisms.
- Integrated the Gemini API to dynamically generate automated exam questions, streamlining the instructor assessment pipeline.
- **Tech Stack:** Next.js, Express.js, PostgreSQL, Prisma ORM, Socket.IO, Gemini API

Smart ID Card Detection - CA-YOLOv8 | github.com/Nithwin/Smart_ID_Card_Detection

- Designed "CA-YOLOv8" - a heavily customized object detection architecture focused entirely on identifying tiny ID cards in live camera feeds.
- Implemented custom attention mechanisms (CBAM) and a small-object detection head to accurately detect partially covered or distant cards.
- Wired it into a complete pipeline that checks for ID cards, identifies the student via InsightFace, and flags violations on a Flask web dashboard.
- **Tech Stack:** Python, Custom YOLOv8, InsightFace, OpenCV, Flask, SQLite

Terminal TCP/IP Chat Network | github.com/Nithwin/Cpp_Projects

- Developed a low-latency, terminal-based chatting application and message broadcaster utilizing core computer networking concepts.
- Implemented raw TCP/IP socket programming within a client-server architecture to ensure reliable, real-time bidirectional data transmission.
- **Tech Stack:** C/C++, Socket Programming, TCP/IP, Linux Networking

ACHIEVEMENTS

- **Competitive Programming:** Solved 750+ Data Structures and Algorithms problems on LeetCode.
- **Innovation Excellence (2x 1st Prize):** Won university competitions for an AI Classroom Behavior Monitoring system and a YOLOv8 ID Card Detection pipeline.